

Distributed ledger technology	Programming languages
Ethereum network	Solidity
Quorum	Solidity
Corda	Kotlin, Java
Hyperledger Fabric	Go, Java, JavaScript

Table 3: Programming languages used for DLT

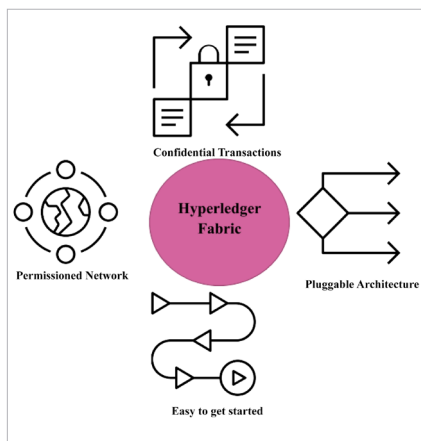


Figure 3: Factors driving Hyperledger Fabric

as a base for developing blockchain products and applications through plug-and-play components mainly meant to target private enterprises.

Hyperledger Fabric is an open source distributed ledger framework launched by the Linux Foundation that provides a versatile client utility based on consensus, privacy and membership services. The speed of a transaction is high when there are a smaller number of nodes.

Consider, for example, a seller who wishes to sell products at different rates to various clients using the same platform. It is imperative that the rates be kept discreet (private) among the several buyers. This is possible using the 'channels' feature of Hyperledger Fabric. For complete isolation of transactions, the 'private data' feature can be explored, where hashes are shared on the ledger as

proof of transaction ('private data' is allowed to be shared amongst 'collection members'). Hyperledger supports membership of specified chains with permission from peers.

As evident from Figure 4, Hyperledger Fabric segregates the transaction process into three parts – smart contracts, transaction ordering, and transaction validation and commitment. This permits scalability across the network and also in the trust and verification levels, thereby increasing the performance efficiency manifold. The plug-and-play concept permits the reusability of existing features to integrate readymade modules in it. Endorsers (simulate, sign or reject transactions), consenters (complete PBFT or practical byzantine fault tolerance protocol on the transactions) and committers (receive the batch of transactions and write to the ledger) ensure the robustness of the system. This is something identical to using design architectures like VGG 16 in deep learning, using transfer learning techniques for our desired applications.

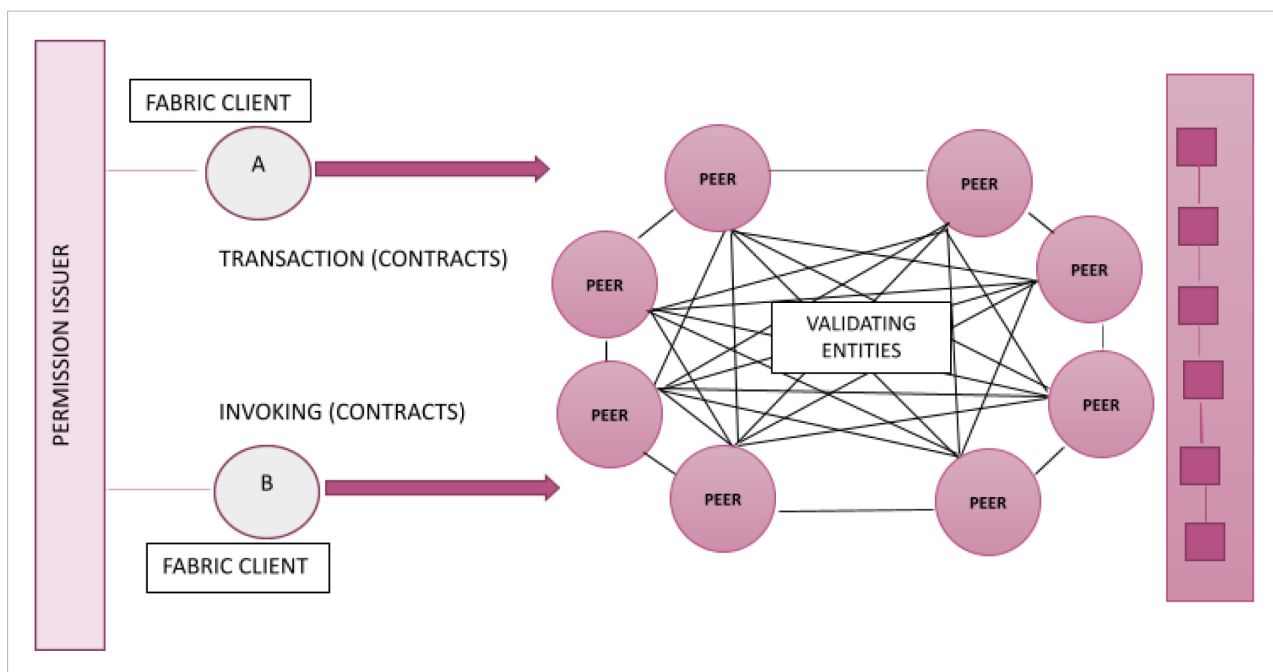


Figure 4: Hyperledger Fabric model